



FACULTY OF COMPUTING AND INFORMATION

TECHNOLOGY Assignment

**BMIT2123 Internet of Things
202505**

Student's name/ ID Number : Ethel Ng Yi Yan / 24WMR08861

Student's name/ ID Number : Gan Khai Li / 24WMR08863

Student's name/ ID Number : Kit Chin Jie Ying / 24WMR08870

Student's name/ ID Number : Chan Jin Wei / 24WMR08851

Programme : Bachelor of Data Science (Honours) - RDS

Tutorial Group : 4

Date of Submission to Tutor : 12/09/2025

Table Content

Table Content	2
1.0 System Title - Smart Living System	3
2.0 Description of System	4
2.1 Objective	4
2.2 Business Value / Advantage	5
2.3 Targeted Users	6
2.4 Why Does Everyone Need Our System?	8
3.0 System Design	10
3.1 Reference System	10
3.2 Block diagram of system architecture	11
3.3 Functional / Module Descriptions	12
3.3.1 Smart Temperature System	12
3.3.2 Smart Entrance System	13
3.3.3 Smart Detection System	14
3.3.4 Smart Control System - User Interface	16
3.4 Snapshot/Picture	19
3.4.1 Smart Temperature System	19
3.4.2 Smart Entrance System	19
3.4.3 Smart Detection System	20
3.4.3.1 Smart Smoke and Flame Alert System	20
3.4.3.2 Smart Lightning System	20
3.4.4 Smart Control System - User Interface	21
4.0 Task Allocation	22
5.0 Lesson that we learned / Problem that we faced	23
5.1 Analysis of data	24
5.1.1 Correlation between temperature and humidity	24
5.1.2 Hourly average of temperature and humidity	25
6.0 Conclusion	26
7.0 Reference	28

1.0 System Title - Smart Living System

The Smart Living System is an advanced, technology-driven solution designed to enhance comfort, security, efficiency and overall quality of life within residential homes. By seamlessly integrating a wide range of smart devices, automation technologies, sensors and real-time data processing capabilities, the system creates a fully connected and intelligent living environment. This allows users to monitor, manage and optimize various aspects of their home, including lighting, climate control, security systems, household appliances and energy consumption, all through a single, easy-to-use platform.

By leveraging the latest advancements in the Internet of Things (IoT) and mobile application control, the Smart Living System delivers a convenient and adaptive experience. It enables the system to automatically adjust settings and perform everyday tasks such as temperature regulation, door access, hazard alerts and lighting control, minimizing manual effort while maximizing convenience.

Users can interact with the system through multiple interfaces such as intuitive mobile applications, voice commands or customizable preset schedules, providing flexible control options tailored to individual needs and lifestyles. Whether remotely adjusting the thermostat before arriving home, receiving real-time security alerts or automating energy-saving measures during peak usage periods, the system empowers users to make smarter decisions that benefit both their household and the environment.

At its core, the Smart Living System transforms a traditional house into a fully integrated smart home ecosystem. It adapts to changing conditions detected by sensors, ensuring every part of the home works together toward safety, convenience and sustainability. This redefines the modern residential experience by setting new standards for intelligent living in today's connected world.

2.0 Description of System

2.1 Objective

The primary objective of the Smart Living System is to significantly enhance the overall quality of life within residential environments by harnessing the power of advanced intelligent automation, seamless connectivity and real-time control. This comprehensive system is meticulously designed to integrate and manage a wide range of home functionalities, including lighting, climate control, security systems, household appliances and energy management, within a single, unified and user-friendly platform. By doing so, it transforms ordinary homes into highly efficient, comfortable and secure living spaces.

At the core of the Smart Living System lies the integration of Internet of Things (IoT) technologies, which enable various smart devices and sensors throughout the home to communicate and operate in harmony. This connectivity is further enhanced by the use of **Firestore cloud services**, which provide secure data storage, real-time synchronization and remote accessibility. Through cloud integration, homeowners can monitor system status, receive alerts and control devices from virtually anywhere using the mobile application.

The system's proactive monitoring capabilities allow it to continuously observe and assess multiple aspects of the home environment, such as temperature fluctuations, lighting levels, security breaches and energy consumption patterns. Based on this real-time data, the Smart Living System can dynamically adjust settings to optimize comfort, safety and efficiency. For example, it can automatically regulate heating and cooling systems to maintain ideal indoor climate conditions while minimizing energy use or it can detect unusual activities and promptly alert homeowners or security services.

Ultimately, the Smart Living System goes beyond traditional home automation by creating a responsive and cloud-connected residence that integrates safety, convenience and energy efficiency. By enhancing comfort, improving security and fostering better energy management, this system redefines modern residential living and sets a new standard for smarter, more sustainable homes of the future.

2.2 Business Value / Advantage

The Smart Living System provides significant value across multiple dimensions, making it a compelling offering for homeowners, property developers and businesses in the smart home ecosystem. There are several key advantages such as :

1. **Energy Efficiency & Cost Savings** - Smart scheduling and automated control of devices like HVAC systems, lights and appliances help reduce energy consumption, translating into lower utility bills and a smaller environmental footprint.
2. **Enhanced Security & Surveillance** - Real-time monitoring through smart cameras, motion detectors, door/window sensors and smart locks ensures 24/7 protection. The system sends instant alerts in case of suspicious activity, fire, gas leaks or intrusions.
3. **Remote Monitoring & Control** - Users can control their home environment from anywhere using a mobile app or web interface. Whether it's adjusting room temperature, locking the door or checking camera feeds, the system offers full access on-the-go.
4. **Improved Property Value** - Smart homes are more attractive to modern buyers and tenants. Integrating such a system can increase the resale value and marketability of the property.
5. **Scalable & Future-Proof** - The modular architecture allows the system to be expanded or upgraded as new smart devices and technologies emerge, ensuring long-term usability.
6. **Sustainability & Eco-Friendliness** - The system promotes responsible energy usage and contributes to global sustainability goals by minimizing waste and carbon emissions.

2.3 Targeted Users

1. Homeowners Looking for Comfort, Control and Efficiency

For today's homeowner, managing daily household tasks can often feel overwhelming and time-consuming. The Smart Living System transforms this experience by automating and simplifying many aspects of home management. With just a tap on a smartphone, users can control lighting, fans and windows or receive real-time alerts for safety concerns. The system helps conserve energy by turning off lights when no one is present and maintaining ideal indoor conditions through automated temperature and humidity control. Beyond convenience, the system enhances security by providing smart door access, motion detection and instant alerts for emergencies such as fires, gas leaks or smoke. Whether at home or away, users gain full control over their living environment, which increases comfort and peace of mind.

2. Busy Professionals Who Want to Automate Daily Routines and Ensure Security While Away

In a fast-paced world, busy professionals often struggle to keep up with household needs due to heavy work schedules and frequent travel. The Smart Living System is designed to meet their needs by providing automated controls that can also be managed remotely through the mobile app. Users can set thresholds for temperature and humidity, automatically adjust lighting based on occupancy and brightness and receive alerts if unusual activity or hazards are detected while they are away. Security is reinforced with RFID-based access, motion detection at the entrance and real-time notifications delivered via Firebase cloud. Overall, the system allows professionals to maintain peace of mind and security without the stress of constantly supervising their home.

3. Families with Children Seeking a Safe and Convenient Living Environment

Safety and convenience are top priorities for families, especially those with young children. The Smart Living System offers intelligent controls to make homes safer and more manageable for family routines. Parents can receive alerts if smoke, gas or fire hazards are detected, ensuring immediate action can be taken. Smart entrance features such as RFID card access and remote door control help ensure children can

safely enter the home after school while preventing unauthorized entry. Automated lighting ensures rooms are well lit when occupied and energy is not wasted when empty. These functions provide families with peace of mind, while also making everyday household management easier.

4. Elderly or People with Disabilities, Who Benefit from Voice Control, Automation and Emergency Alerts

For elderly individuals and people with disabilities, maintaining independence and safety at home can be challenging. The Smart Living System significantly improves quality of life by automating essential tasks such as lighting control, temperature regulation and secure door access. Crucially, the system includes safety features like smoke, flame and gas detection, which automatically trigger alarms, ventilation and notifications to caregivers or family members. Remote app control also ensures that help can be provided quickly when needed. By reducing physical effort and improving safety, the system promotes independence and reassurance for users and their loved ones.

5. Smart Home Enthusiasts Who Want to Explore the Full Potential of Home Automation

Smart home enthusiasts enjoy experimenting with technology and automation. The Smart Living System provides a solid platform for exploring sensor-based control, real-time monitoring and cloud-connected management. Enthusiasts can create practical automation scenarios, such as automatically turning on lights when a room is occupied, opening windows when the temperature rises or triggering alarms and water pumps in response to hazards. With Firebase cloud integration, they can store and access home data remotely, while alerts and notifications are displayed directly on the user interface for immediate awareness. This system offers enthusiasts reliable, functional automation that demonstrates the practical power of IoT in everyday living.

6. Property Developers & Real Estate Firms Aiming to Build or Retrofit Homes with Modern, High-Tech Appeal

In real estate, incorporating smart technology into homes has become a key selling point. Property developers and real estate firms can use the Smart Living System to

enhance the attractiveness and value of their properties. Whether building new developments or retrofitting existing buildings, adding smart sensors and automation features appeals to modern buyers and renters who prioritize convenience, sustainability and safety. Features such as automated lighting, smart entrance access and hazard detection systems reduce operational concerns and make homes more appealing. Offering smart-enabled homes helps developers meet demand for intelligent living spaces and increases property marketability.

7. Rental Property Managers (e.g., Airbnb Hosts) Wanting Remote Access, Security and Energy Control for Multiple Units

Managing rental properties comes with unique challenges, especially with frequent guest turnover. The Smart Living System provides an effective solution by offering remote access control, real-time monitoring and automated environmental management. Smart locks with RFID access eliminate the need for physical key exchanges, while motion sensors and entrance cameras enhance security for both hosts and guests. Automated lighting and ventilation control help conserve energy when units are unoccupied. Security and safety alerts are displayed directly on the user interface, keeping managers informed of potential issues in real time. These features not only improve guest comfort but also make property management more efficient, reliable and cost-effective.

2.4 Why Does Everyone Need Our System?

In today's fast-paced, technology-driven world, homes need to be more than just traditional living spaces. They must evolve to meet growing demands for convenience, safety, efficiency and sustainability. The Smart Living System addresses this by transforming ordinary houses into intelligent, connected environments. It provides a central platform that allows users to manage and automate key aspects of their home. Whether it is controlling lighting, adjusting temperature, scheduling appliances or monitoring security, everything can be done easily through a smartphone or tablet. This gives users greater comfort and control, while making daily routines smoother and more efficient.

Safety is a major reason why this system is essential. It integrates smart surveillance cameras, motion detectors, smoke and gas leak sensors and environmental monitors to

provide round-the-clock protection. If unusual activity or emergencies occur, alerts are displayed instantly on the user interface so residents can respond quickly. This ensures peace of mind whether they are at home or away.

The Smart Living System also helps reduce unnecessary energy use by managing consumption intelligently. Lights can switch off automatically in empty rooms, thermostats can maintain optimal temperatures and appliances can be scheduled to run only when needed. These functions lower utility bills while supporting sustainable living practices.

For people with busy lifestyles or limited mobility, the system's automation and easy controls are especially valuable. It can handle everyday tasks such as turning off lights at night, locking doors or starting appliances at set times without requiring manual effort. Remote access through the cloud lets users check and manage their home even when traveling, while the simple user interface ensures that the system remains accessible and user-friendly for people of all ages.

Ultimately, the Smart Living System is not just a convenience as it is a necessity for modern living. It improves comfort, strengthens safety, saves money and promotes sustainability. As homes continue to get smarter, this system ensures that individuals and families remain connected, secure and in control in a constantly changing world.

3.0 System Design

3.1 Reference System

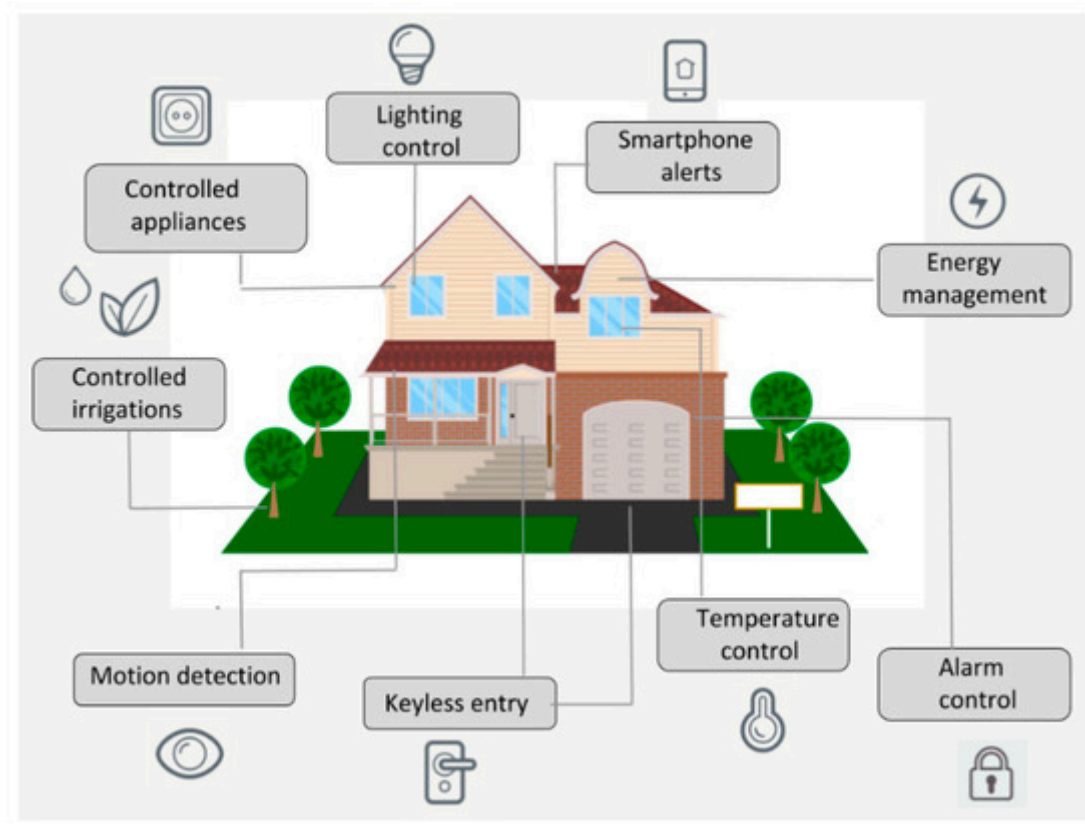


Figure 3.1: An IoT-based smart home depicting the use of smart sensing devices for different purposes

Stolojescu-Crisan et al. (2021) proposed an IoT-based smart home automation system aimed at enhancing comfort, security and energy efficiency in residential environments. The system architecture integrates various sensors and actuators connected to a central microcontroller such as Arduino or Raspberry Pi, which communicates data using wireless protocols like Wi-Fi and Bluetooth. This architecture supports real-time monitoring and remote control of household devices via a mobile or web-based graphical interface.

Key functionalities include environmental monitoring such as temperature, humidity and gases, smart lighting control, motion detection and door or window status monitoring. The system also supports automation rules and scheduling, enabling context-aware device operation such as activating lights upon motion detection at night.

Collected data is processed centrally and visualized for the user, supporting intuitive interaction and control. The system emphasizes low cost, scalability and modularity, allowing

easy integration of additional components. With a strong focus on remote accessibility and user-defined automation, this solution exemplifies a practical model for smart living applications.

3.2 Block diagram of system architecture

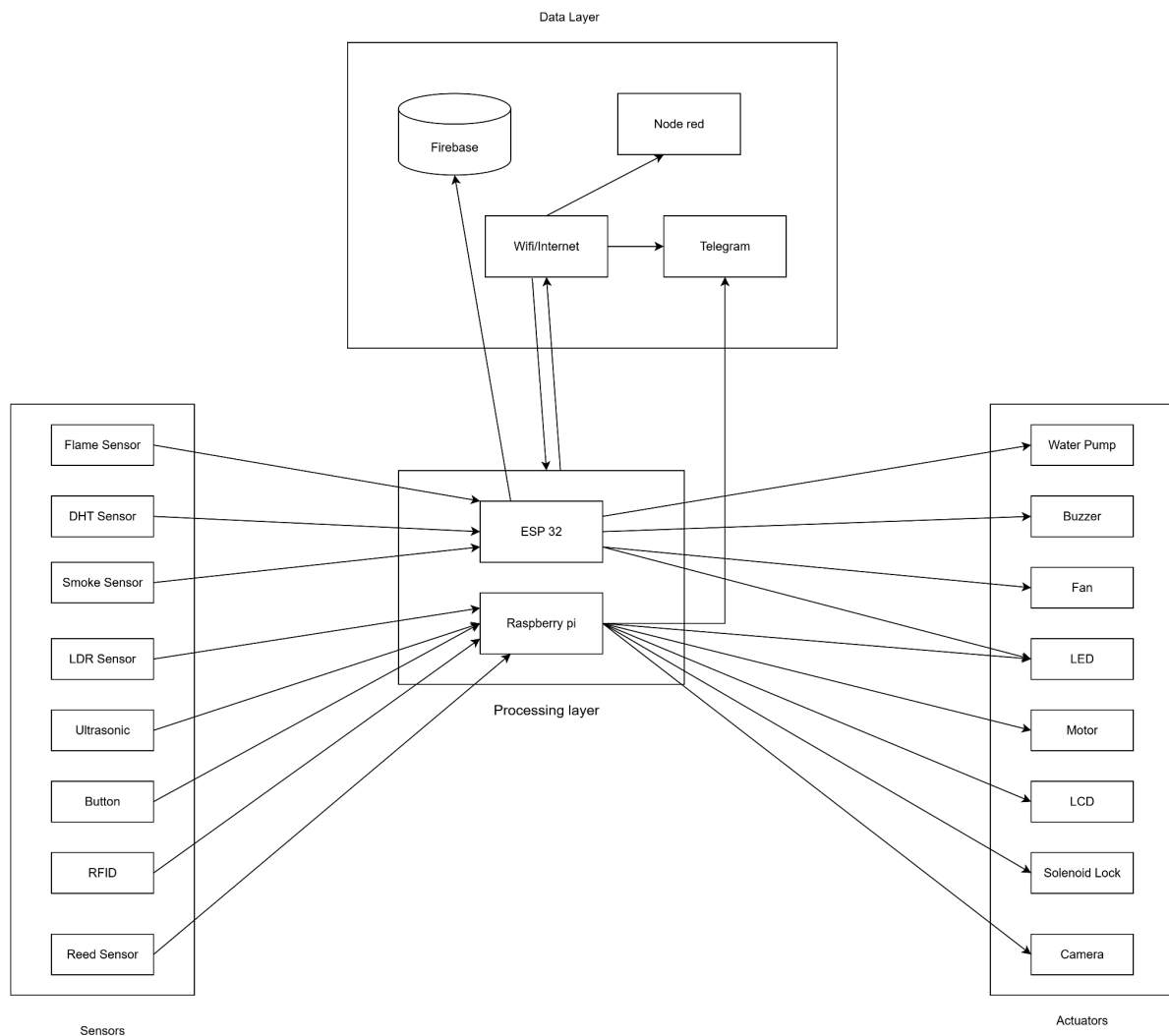


Figure 3.2: Block Diagram of System Architecture

Table 3.1: Component Used in Each System

System	Sensors	Actuators
Smart Temperature System	DHT Sensor Ultrasonic Sensor	Motor Fan LCD
Smart Entrance System	Ultrasonic Sensor Button RFID	LCD Buzzer Solenoid Lock

	Reed Sensor	Camera
Smart Detection System - Smart Smoke and Flame Alert Module	Flame Sensor Smoke Sensor (MQ-2)	Buzzer LED Water Pump
Smart Detection System - Smart Lightning System	Ultrasonic Sensor LDR Sensor	LED

3.3 Functional / Module Descriptions

3.3.1 Smart Temperature System

The smart temperature system monitors and adjusts indoor environmental conditions to maintain comfort and improve energy efficiency. It integrates sensor input, automatic controls and remote access via app.

1. Environmental Sensing with DHT21 Sensor

- The DHT21 sensor continuously measures indoor temperature and humidity levels. These measurements are used by the system to make real-time adjustments for optimal comfort.

2. Automatic Fan and Window Control

- When the indoor temperature exceeds a preset threshold, the system automatically opens the window and turns on the fan to cool the space. If humidity rises above a defined limit, the system closes the window to prevent excess moisture from entering.

3. Ultrasonic Occupancy Detection

- An ultrasonic sensor detects whether anyone is present indoors. If no occupants are detected, the system automatically closes the window to maintain security and improve energy efficiency.

4. App Remote Control

- The homeowner can manually open or close the window and control the fan remotely through the app, allowing for convenient adjustments from anywhere.

3.3.2 Smart Entrance System

The smart entrance system integrates multiple technologies to enhance both security and convenience. It combines automated controls with remote access to provide a seamless entry experience.

1. RFID Card Access

- Authorized users can unlock the door by scanning a registered RFID card. Once verified, the door automatically unlocks, granting entry.

2. Camera and Notification

- A built-in camera with CCTV functionality is embedded in the door. When a visitor presses the doorbell, the camera activates and begins live video streaming. At the same time, a notification is sent to the homeowner's app, enabling real-time monitoring of visitors.

3. Ultrasonic Motion Detection

- An ultrasonic sensor detects motion near the entrance. When movement is detected, the system can trigger alerts or activate the camera to record, providing an additional layer of security.

4. App Remote Control

- The homeowner can remotely lock or unlock the door through an app. This allows them to grant access to visitors without being physically present, offering both convenience and flexibility.

3.3.3 Smart Detection System

The smart detection system enhances home safety and energy efficiency through fire, gas and lighting control. It consists of two main modules: smoke detection and smart light detection.

A. Smart Smoke and Flame Alert Module

1. Gas and Flame Sensing

- The system is equipped with gas sensors (MQ2) and flame sensors to continuously monitor the environment for smoke, fire and gas leaks such as methane and alcohol. These sensors are sensitive to ensure that changes of air quality are detected properly. If readings exceed defined safety thresholds, it automatically initiates emergency responses.

2. Automatic Safety Response

- When danger conditions such as a potential fire, smoke or gas hazard is detected, the system triggers a buzzer alarm and LED light to alert occupants. At the same time, the system will activate the water pump to help extinguish flames before the flame spreads further. The windows will be opened for ventilation and unlocks the door to provide a safe exit route.

3. App Notification

- The system integrates with a mobile application that ensures homeowners are notified instantly, even if they are away from home. Once a hazard is detected, the app sends a real-time push notification to alert the homeowners. This feature allows homeowners to stay informed and take immediate action, such as contacting emergency services or checking on family members.

4. App Control

- The mobile app gives homeowners direct control over key safety devices, including the active buzzer, LED light and water pump. Homeowners can turn these components on or off remotely, depending on the situation. For example, the buzzer can be silenced after

confirming safety, the LED can be used as a status signal and the water pump can be manually activated for extra fire suppression.

B. Smart Lightning Module

1. Automatic Light Control

- An occupancy sensor detects whether people are present in a room. Based on detection, the system automatically switches LED lights on or off to save energy.

2. Ambient Light Detection

- In addition to occupancy sensing, the system is equipped with an Light Dependent Resistor (LDR) sensor that continuously measures the surrounding brightness. If the environment is dark, the lights will automatically switch on, even if occupancy is detected. Conversely, if the room already has sufficient natural lighting, the system keeps the lights off to save energy. This can balance comfort and energy efficiency.

3. App Control

- Through the mobile app, homeowners can manually control the lights at any time, regardless of sensor readings. They can turn lights on or off remotely or override automatic settings when specific lighting conditions are preferred whenever they want.

3.3.4 Smart Control System - User Interface

The smart control system user interface provides a centralized platform for monitoring and controlling all connected smart modules. Developed using Node-RED, the interface ensures real-time visibility, manual override and efficient interaction through a user-friendly dashboard.

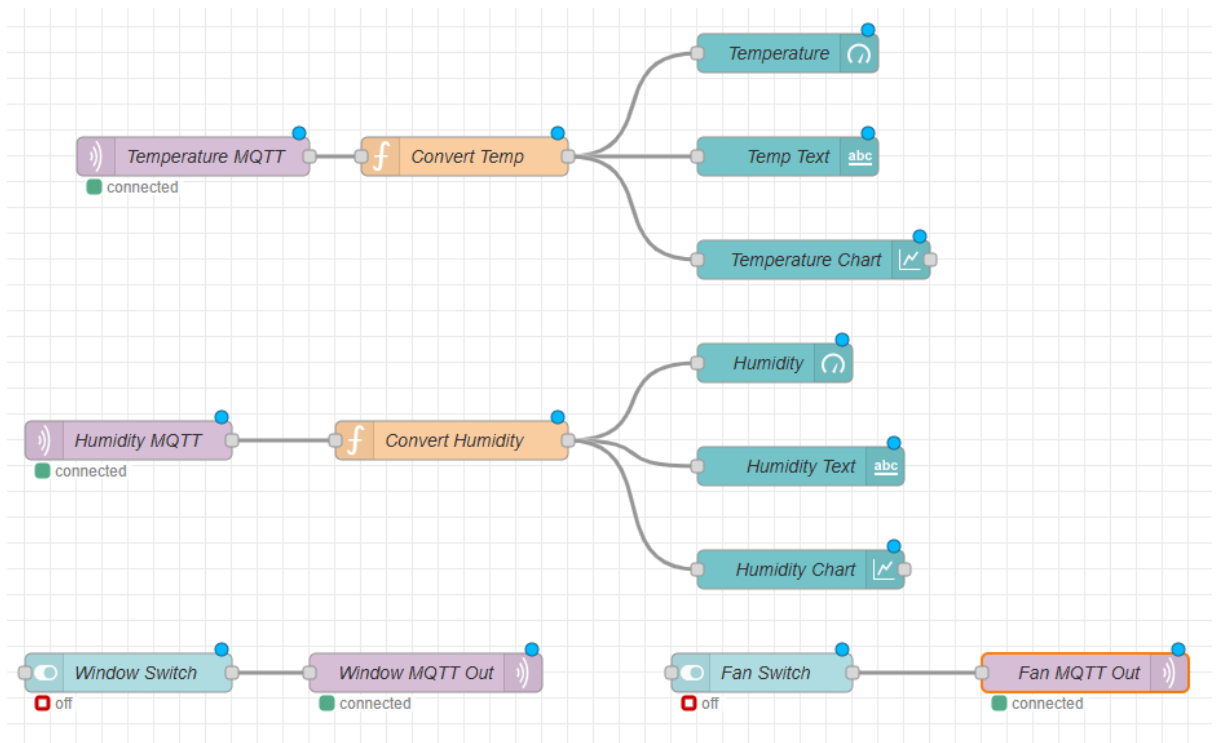


Figure 3.3.4.1: Node-RED Design of Smart Temperature System

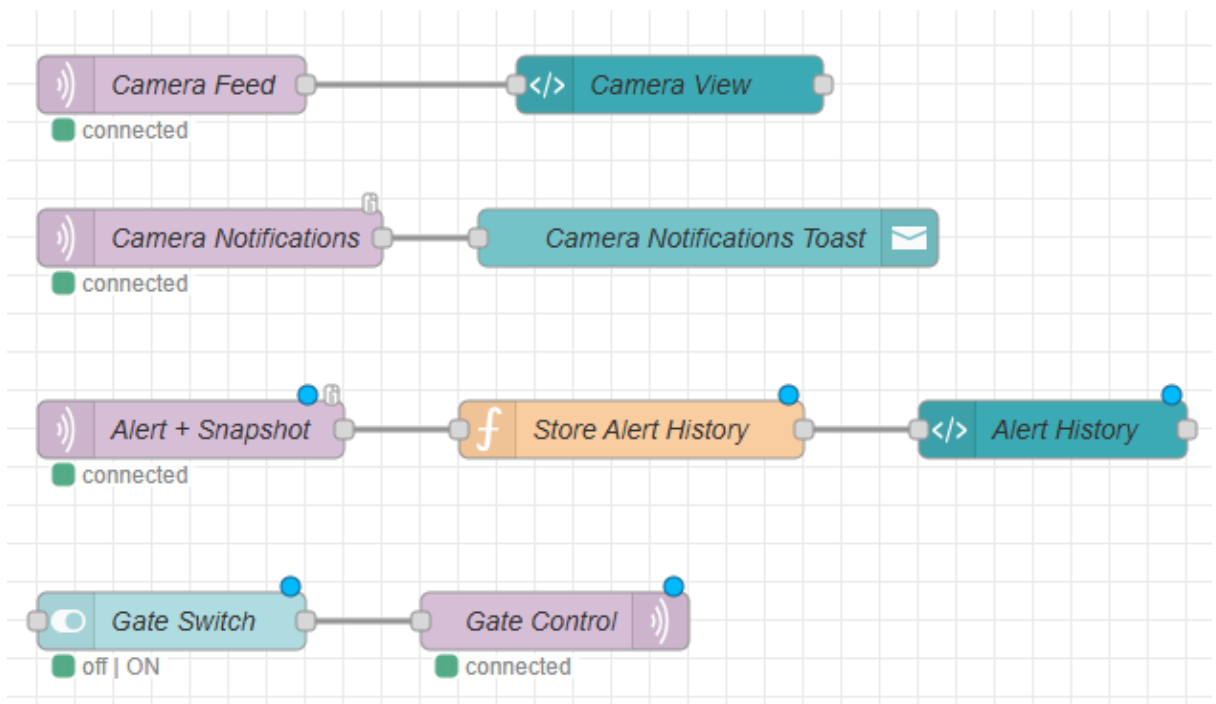


Figure 3.3.4.2: Node-RED Design of Smart Entrance System

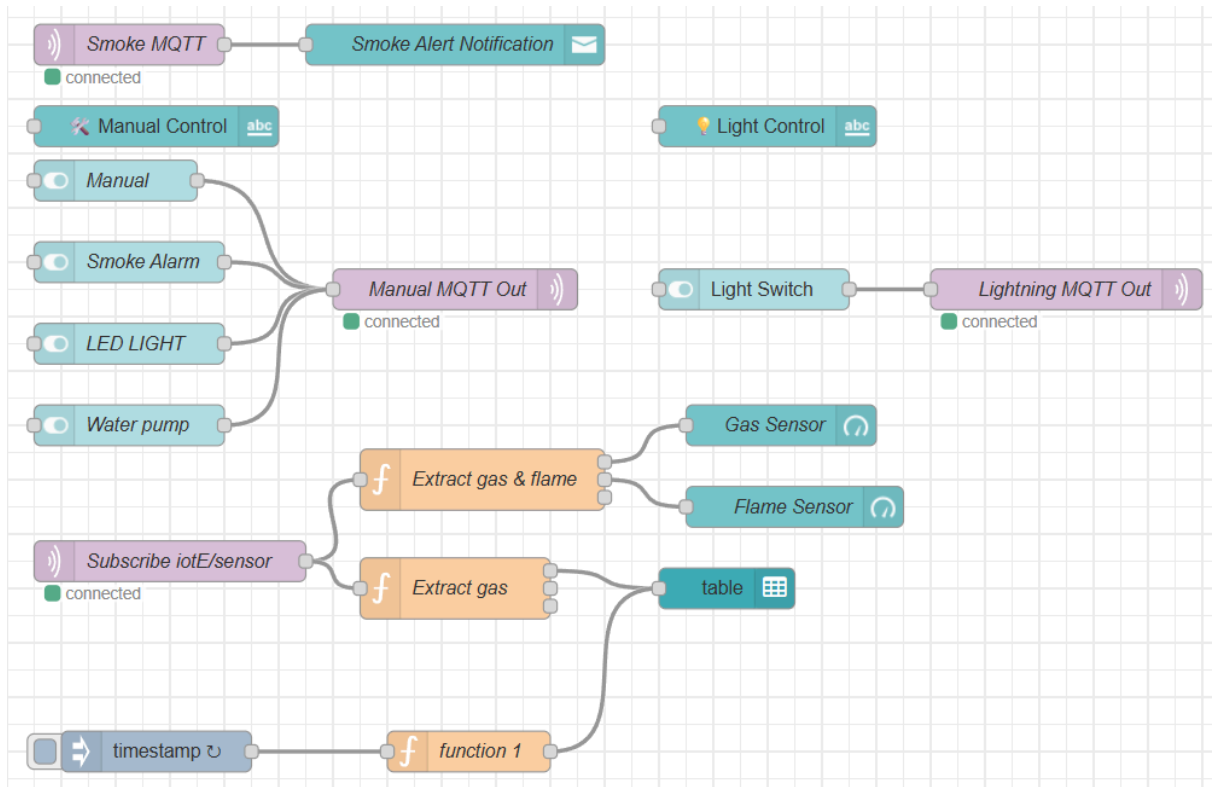


Figure 3.3.4.3: Node-RED Design of Smart Detection System

1. Real-Time Monitoring

- The interface provides continuous live data from various sensors, including temperature, humidity, motion, smoke and flame detection. Data is presented in real-time through clear text indicators, intuitive icons and interactive charts for easy understanding.

2. Live Camera Feed

- A live video stream from the entrance camera is seamlessly integrated into the dashboard, enabling users to visually monitor visitors and entrance activities directly from the interface.

3. Manual Device Control

- Users can manually control devices such as the gate, fan, window and LED lights via intuitive toggle switches. This functionality allows overriding automatic system settings, granting full individual control over each device.

4. Alert Notification and Snapshot Display

- When critical alerts such as fire detection, gas leaks or motion near the entrance are triggered, the system immediately notifies the user and

captures snapshots to provide visual context, facilitating swift assessment and response.

5. Alert History Logging

- All alerts, along with their associated snapshots and timestamps, are stored in a dedicated alert history panel. This feature enables users to review past incidents and analyze patterns for enhanced security and safety.

6. Remote Access via App

- The dashboard is mobile-responsive and accessible via app, enabling users to monitor and control the system remotely from smartphones or other internet-connected devices.

7. Module-Based Interface Layout

- The interface is thoughtfully organized into distinct modules for each subsystem, such as entrance control, environmental monitoring and smoke detection, which makes navigation intuitive and interactions efficient.

3.4 Snapshot/Picture

3.4.1 Smart Temperature System

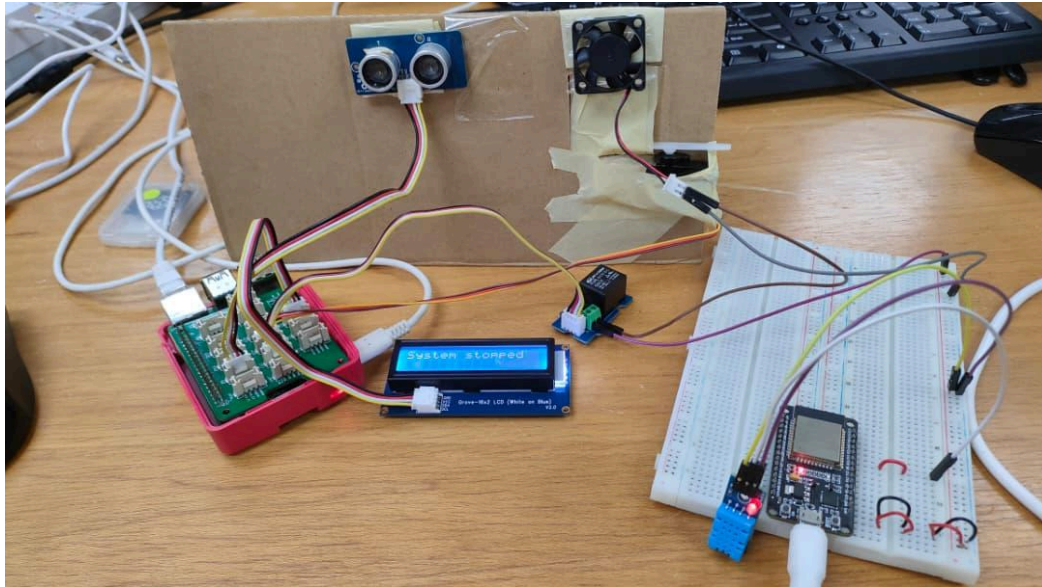


Figure 3.4.1 Smart Temperature System

3.4.2 Smart Entrance System

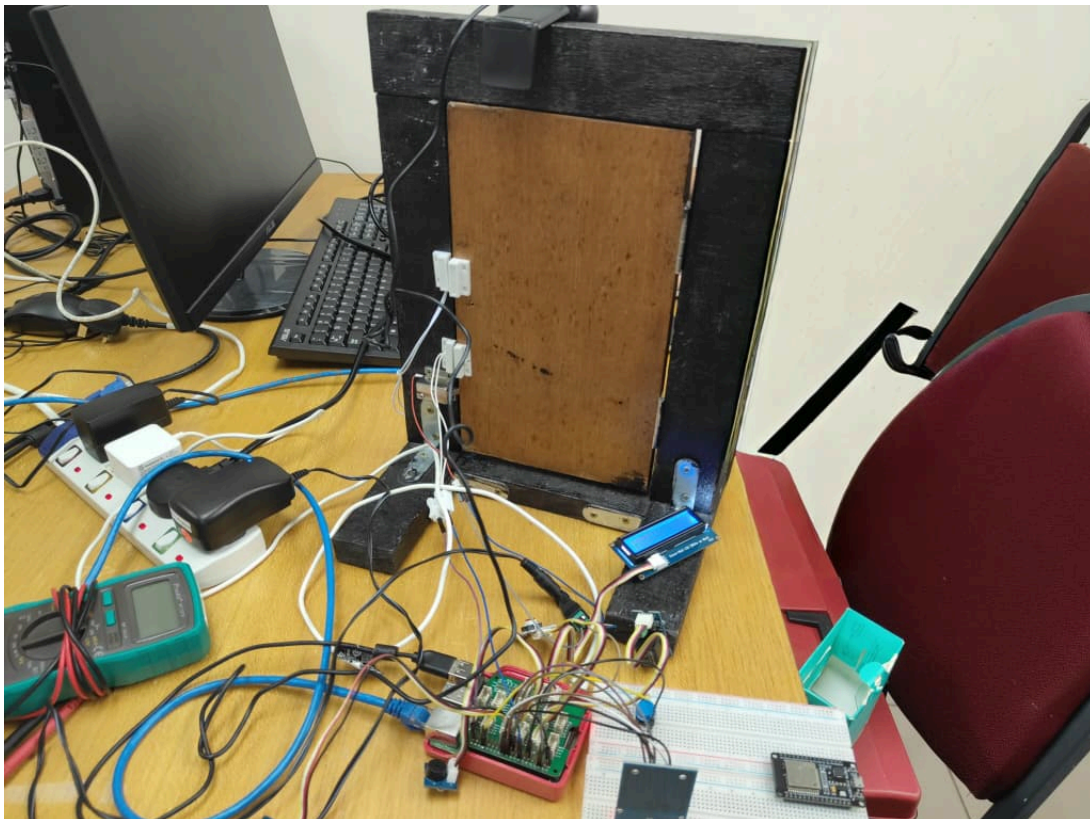


Figure 3.4.2 Smart Entrance System

3.4.3 Smart Detection System

3.4.3.1 Smart Smoke and Flame Alert Module

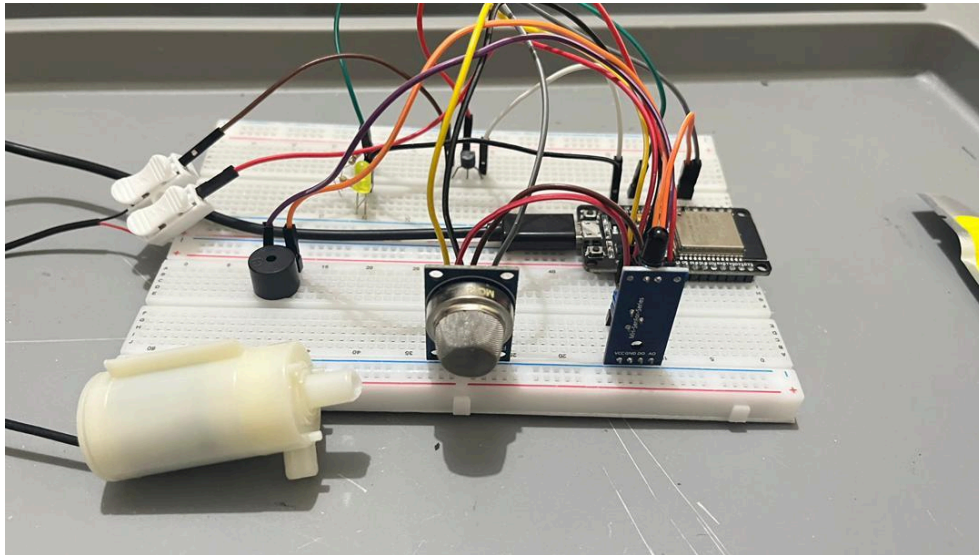


Figure 3.4.3.1 Smart Smoke and Flame Alert Module

3.4.3.2 Smart Lightning Module

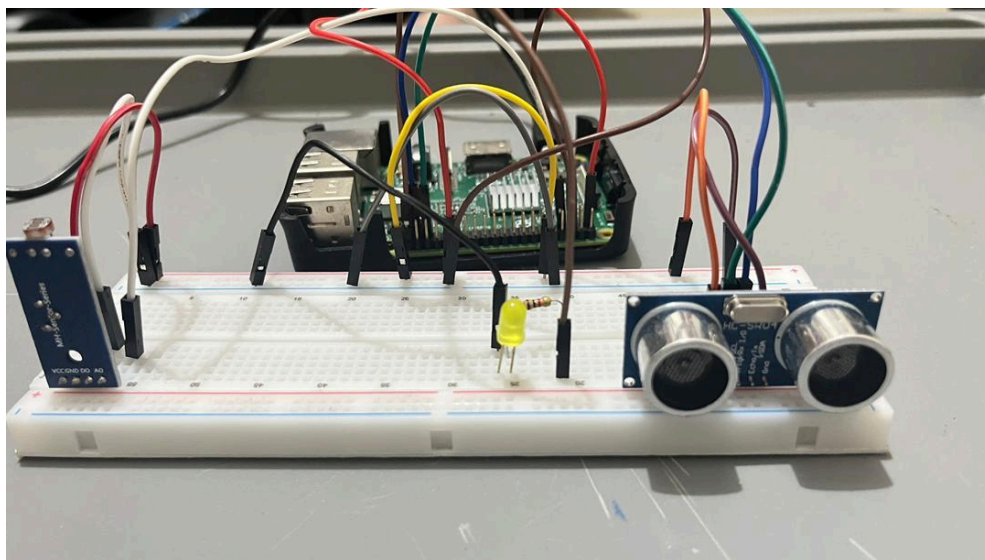


Figure 3.4.3.2 Smart Lightning Module

3.4.4 Smart Control System - User Interface

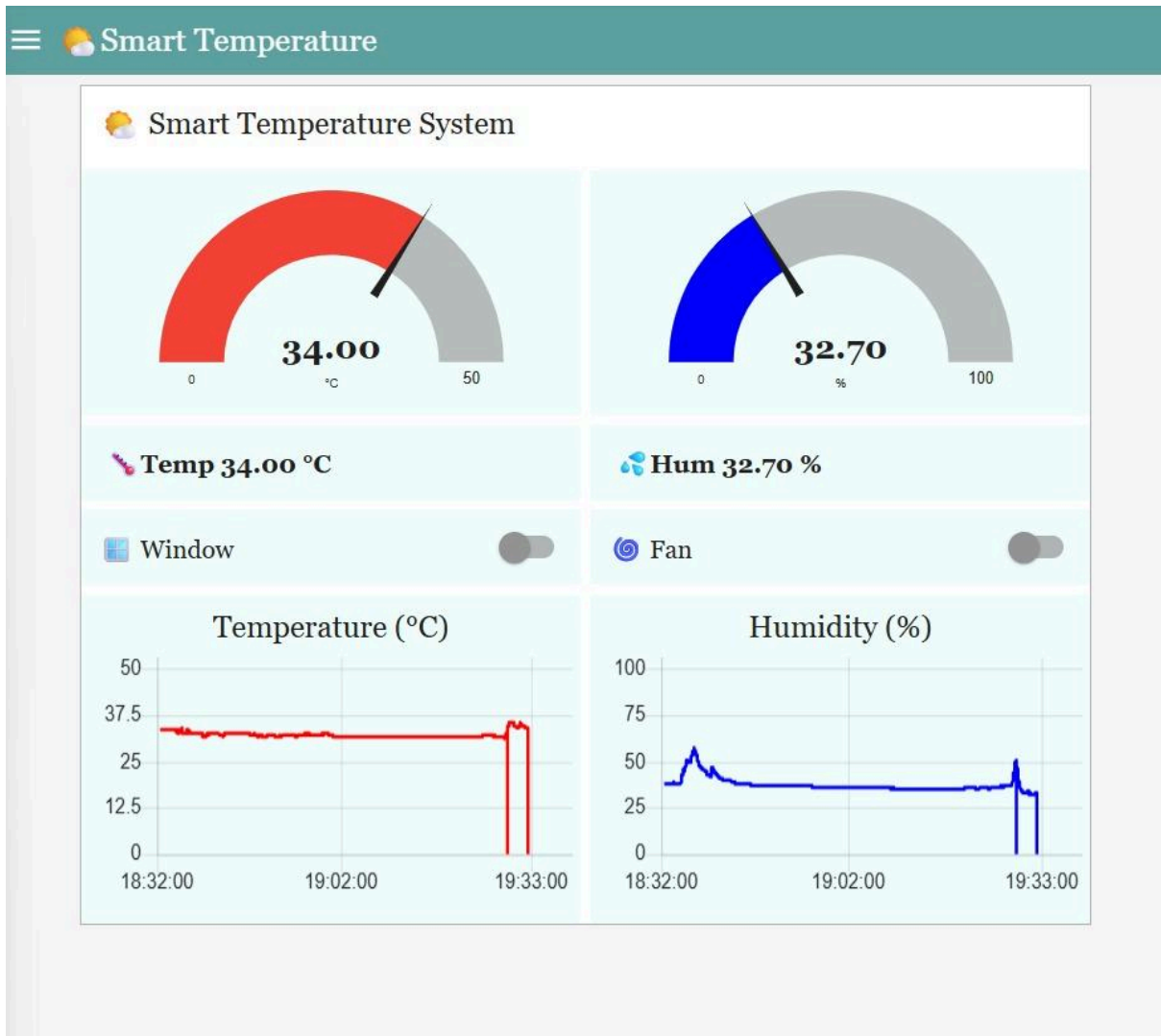


Figure 3.4.4.1: User Interface of Smart Temperature System

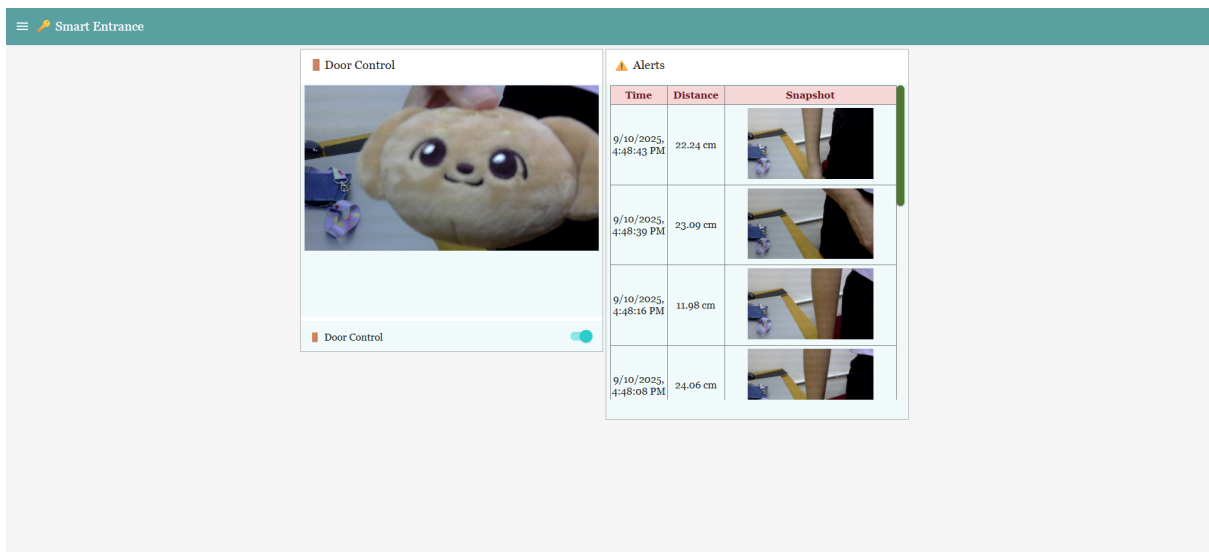


Figure 3.4.4.2: User Interface of Smart Entrance System

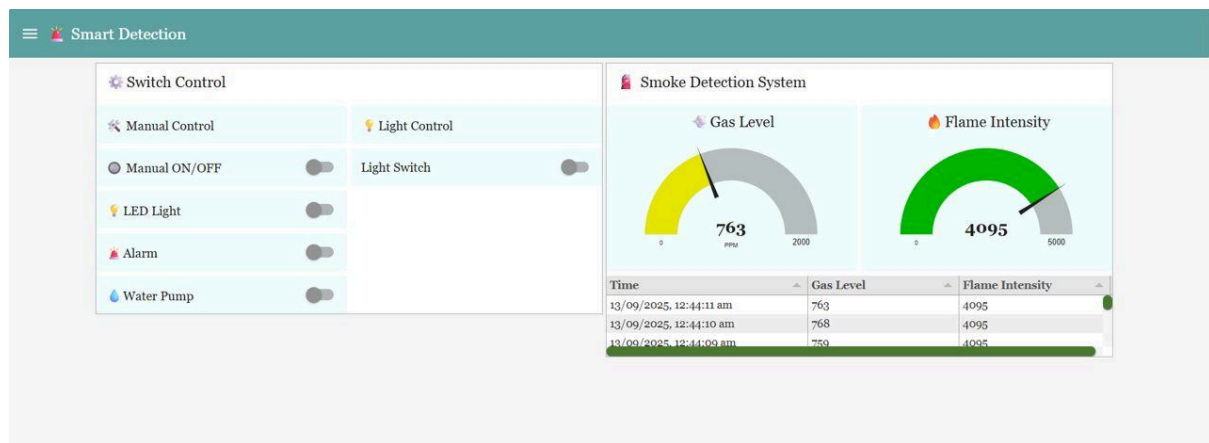


Figure 3.4.4.3: User Interface of Smart Detection System

4.0 Task Allocation

Module Name	Handled by
Smart Temperature System	Chan Jin Wei
Smart Entrance System	Kit Chin Jie Ying
Smart Detection System	Ethel Ng Yi Yan
Smart Control System	Gan Khai Li

5.0 Lesson that we learned / Problem that we faced

Problems that We Faced

During the development and testing of the smart living system, several issues were encountered. Sensor readings such as temperature, humidity and motion were sometimes unstable due to environmental interference, which reduced measurement accuracy. The integration of multiple modules also created conflicts between automatic control and manual override, resulting in occasional unexpected device behavior. Continuous operation of sensors, live camera streaming and communication modules contributed to high power consumption. Another problem was the initial design of the Node-RED dashboard, which appeared cluttered and made monitoring and control less user-friendly. In addition, unstable internet connectivity sometimes disrupted remote access through the app, limiting usability. Finally, security risks were identified in remote access and RFID unlocking, raising concerns about potential unauthorized entry and data privacy.

Solutions to the Problems

During the development and testing of the smart living system, several solutions were applied to address the challenges. To improve sensor accuracy, higher precision sensors and filtering techniques were used to reduce environmental interference. A priority logic was designed in Node-RED to avoid conflicts between automatic and manual control, ensuring stable operation. To manage power consumption, low-power sensors and sleep modes were applied, while event-driven activation helped reduce unnecessary energy use. The Node-RED dashboard was redesigned with a clear layout and module-based organization, making monitoring and control more user-friendly. For unstable internet connectivity, local control and automatic reconnection mechanisms were added to maintain system reliability. Finally, data encryption, dual verification for RFID unlocking and regular firmware updates were introduced to strengthen security and protect user privacy.

Lessons that We Learned

From these challenges, important lessons were gained to improve system design and performance. It became clear that sensor calibration and data filtering are essential for stable and reliable measurements. A structured system architecture with clear priority rules must be implemented to avoid conflicts, ensuring that safety-related functions always take

precedence. Power efficiency can be improved by optimizing sensor polling intervals and enabling low-power modes for inactive modules. The dashboard design should follow a modular and simplified layout, allowing users to monitor and control devices more intuitively. To address connectivity issues, offline fallback mechanisms such as manual switches are necessary to ensure continuous usability. Finally, strong encryption and authentication protocols must be applied to secure remote access and protect user privacy, ensuring both reliability and safety of the smart home system.

5.1 Analysis of data

The Smart Living System is further enhanced by its integration with Firebase, which serves as the cloud backbone for storing and retrieving home activity data. Information collected from sensors and devices such as temperature and humidity readings is securely stored in Firebase for real-time access. By analyzing this data, the system can present users with clear insights on their home environment.

5.1.1 Correlation between temperature and humidity

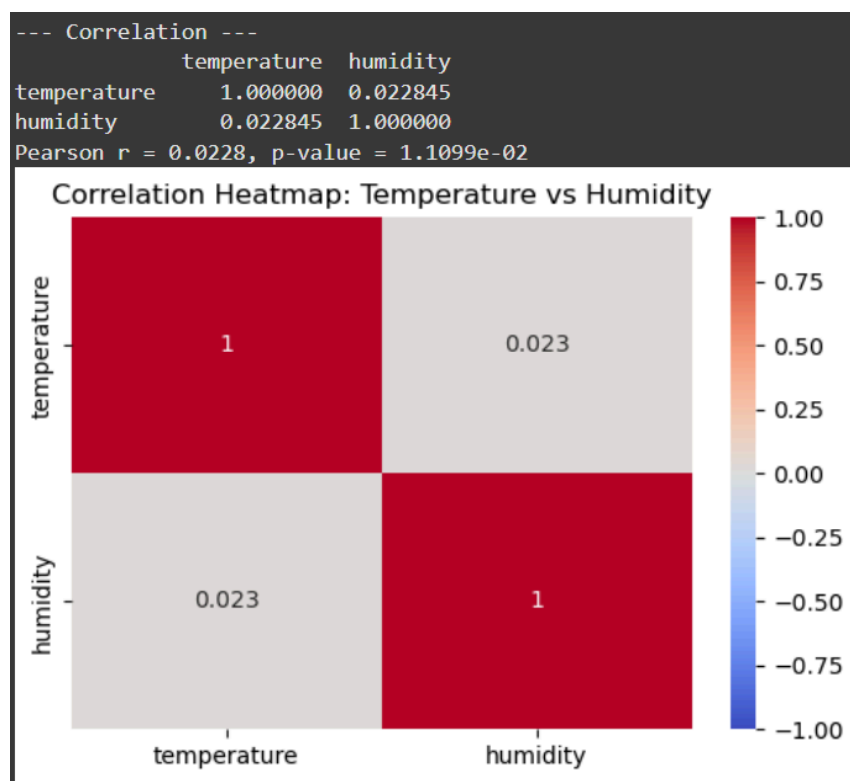


Figure 5.1.1.1: Correlation between temperature and humidity

The correlation heatmap graph representing the relationship between temperature and humidity reveals that the Pearson correlation coefficient is approximately 0.0228, indicating

an extremely weak linear relationship between the two variables. Despite this very low correlation value, the p-value of around 0.011 suggests that this weak association is statistically significant, likely due to a large sample size. The correlation heatmap visually supports this finding, showing almost no color intensity between temperature and humidity, which confirms the near absence of a linear connection. In practical terms, this means that changes in temperature are not meaningfully associated with changes in humidity in this dataset. Thus, temperature and humidity appear to vary independently from each other in this context.

5.1.2 Hourly average of temperature and humidity

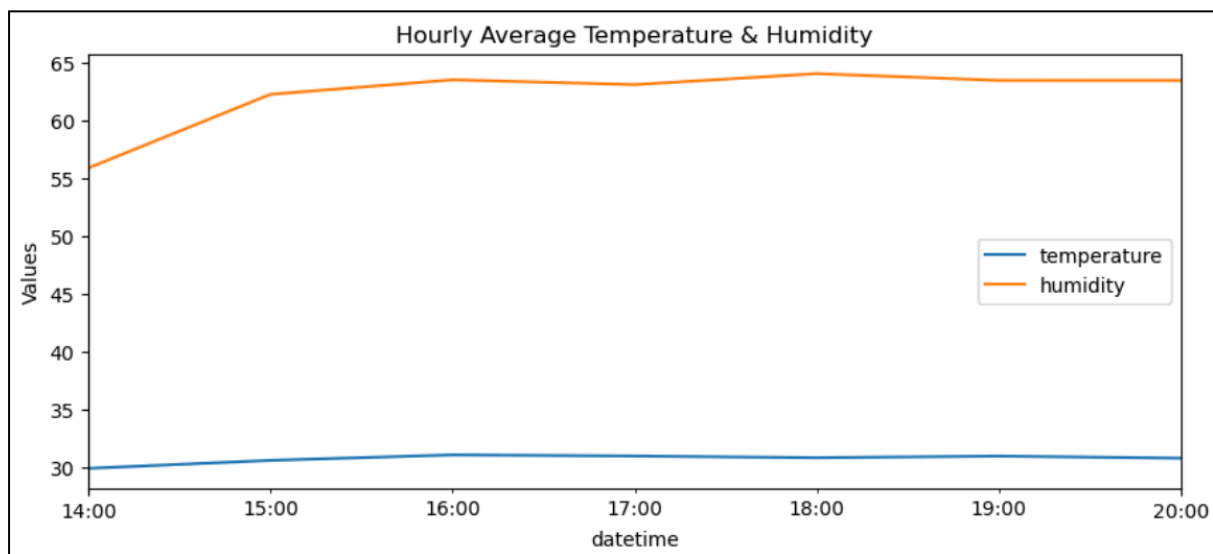


Figure 5.1.2.1: Hourly Average Temperature and Humidity

The line graph represents the hourly average of temperature and humidity over a period from 14:00 to 20:00. During this time, the temperature shows a slight upward trend initially, rising from approximately 30°C at 14:00 to just over 31°C around 16:00. After reaching this point, the temperature remains relatively stable, with only minimal fluctuations observed until 20:00. In contrast, the humidity experiences a more significant change. It begins at around 56% and increases steadily, peaking at approximately 64% between 16:00 and 18:00. From then on, humidity levels stabilize and remain consistent through to 20:00. This pattern indicates that while the temperature does not vary greatly, the amount of moisture in the air increases considerably, likely due to changing atmospheric conditions in the late afternoon. The rising humidity, combined with steady temperatures, may result in a more humid and uncomfortable environment as the day progresses.

6.0 Conclusion

The Smart Living System illustrates the potential of integrating modern technologies such as the Internet of Things (IoT), cloud computing and automation into everyday residential environments. By seamlessly combining intelligent sensing, responsive control and mobile connectivity, it transforms a traditional house into a connected, adaptive and intelligent home. The system is designed to enhance not only safety but also efficiency, convenience and sustainability, providing homeowners with a more reliable and user-centered living experience.

Key features of the system include smart temperature regulation for energy-efficient climate control, secure entrance management using RFID and remote verification and intelligent detection modules such as smoke, flame, gas and motion sensors that ensure real-time monitoring and rapid response during emergencies. A centralized dashboard interface built on Node-RED allows users to easily monitor, control and customize their smart home environment, while the companion mobile app provides remote access and manual override functions for greater flexibility.

During the development and testing process, several challenges were identified. Issues such as unstable sensor readings, conflicts between automatic and manual control, high power consumption, cluttered dashboard design and unstable internet connectivity highlighted the complexity of building a truly reliable and user-friendly smart home system. These challenges provided valuable opportunities to refine the design. Lessons learned emphasized the importance of sensor calibration and data filtering for accurate readings, structured system architecture with clear priority rules to resolve conflicts and power optimization strategies such as low-power modes and event-driven activation. Furthermore, the experience underscored the need for user-friendly dashboard layouts, offline fallback mechanisms like manual switches to ensure reliability and robust security measures including data encryption and dual authentication to protect user privacy.

The Smart Living System ultimately represents more than just convenience, it also demonstrates how technology can enhance comfort, strengthen home security and support sustainable living practices. By reducing wasted energy, improving safety responses and enabling real-time control, it aligns with modern demands for environmentally responsible

and user-centered solutions. Its flexibility also makes it suitable for a wide range of users, including families seeking security, professionals desiring convenience, seniors needing accessible support and property developers looking to integrate smart features into modern housing projects.

With ongoing improvements, scalability and integration of future technologies, the Smart Living System has the potential to set new standards for intelligent, sustainable and inclusive homes. It reflects a vision of modern living where homes are not only places of comfort but also adaptive environments that actively contribute to safety, energy efficiency and overall quality of life.

7.0 Reference

Puzhevich, V. (2019, July 25). *Internet of Things (IoT) in Smart Home: Benefits and Uses* | SCAND Blog. SCAND. <https://scand.com/company/blog/internet-of-things-in-smart-home/>

Stoloiescu-Crisan, C., Crisan, C., & Butunoi, B.-P. (2021). An IoT-Based Smart Home Automation System. *Sensors*, 21(11), 3784. <https://doi.org/10.3390/s21113784>